

NIKHIL NAIR

Kochi, India

Willing to Relocate

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Experience

Super FabLab Kerala – MIT & KSUM [🔗](#)

Nov 2025 – Present

Technology Fellow

Kochi, India

- Developed an **ESP32 Web-Plotter**; offloaded **G-code parsing** to the client-side **JS** engine to reduce MCU load by **60%** while achieving **0.1mm positional precision**.
- Implemented an **Computer vision pipeline** for **automated bed calibration**; engineered real-time object localization to achieve **0.2mm alignment precision** and reduce manual setup overhead by **80%** on the embedded platform.
- Slashed unit cost to **INR 2,000**; **95%** cheaper than industrial alternatives (**~INR 40,000**), saving **INR 10L+** at scale.
- Engineered **bLoom**, an **Edge AI** kinetic installation for **Huddle Global 2025**; optimized **Python/CV** logic to slash CPU overhead by **45%** and achieved **sub-50ms latency** via **Shell-based** headless automation.

TATA Elxsi (Transport Business Unit) [🔗](#)

Oct 2023 – Dec 2023

AI and Robotics Intern

Chennai, India

- Improved obstacle detection by fusing 3D LiDAR with 2D vision using **Python**, extending range in tough conditions.
- Optimized KITTI dataset preprocessing by **~30%** using **ROS2** and **OpenCV**, enhancing the perception pipeline.

Education

Vellore Institute of Technology - Chennai

Sept 2021 - Sept 2025

B.Tech in Computer Science and Engineering (Cyber Physical Systems)

CGPA: 8.21/10 or 3.3/4

- Relevant Coursework:** Data Structures and Algorithms, Operating Systems, Computer Networks, Software Development Lifecycle, Socket Programming, Artificial Intelligence, Core Java, Control Systems, UI/UX, LLMs.

Projects

Urumi V2 — Camera-Assisted 4 Axis CNC Cutter [🔗](#)

(Collaboration with MIT) 2026

- Conducting R&D in collaboration with **Massachusetts Institute of Technology** on Urumi V2, an enhanced CNC cutting machine; leading **camera module** integration and **sensor fusion** while architecting a **lightweight, modular UI** and streamlined **work pipeline**, developing algorithms for **camera calibration** and automated material alignment to enable precise, vision-guided cutting.

NAS-Optimized Deep Learning Model for Concrete Strength Prediction [🔗](#)

(Thesis Project) 2025

- Engineered a **Neural Architecture Search**-optimized model in **Python & Keras** for concrete strength prediction; improved accuracy by **~10%** ($R^2 > 0.90$). Supported design of stronger, cost-effective, and sustainable mixes.

Technical Skills

Languages: Java, Python, C++, Rust, SQL, PHP, HTML/CSS/JS, React.js, Node.js

Frameworks/Tech: Spring Boot, Docker, Kubernetes, Flask, TensorFlow, REST API, OpenCV, ROS2, Jetpack, Qiskit

Tools/Platforms: Git, GitHub, Google Cloud Platform, Arduino, Overleaf, Figma, CI/CD, Kubernetes

Soft Skills: Communication, Team Collaboration, Problem Solving, Presentation, Analytical Thinking, Creativity

Patent & Publication

A System to Recognize Number Plate and Allow Entry of a Vehicle [🔗](#)

2024

Patent Application Number: 202441051527

Published

Comprehensive Review of Blockchain-Oriented Methods in Agricultural SCM [🔗](#)

2026

Harvesting Data: Blockchain, AI and Advanced Innovations in Agriculture

Accepted

Neural Architecture Search & Bio-Inspired Optimization for Concrete Strength Prediction

2026

AI for Civil Engineering Journal – Springer Nature

Review

Inverter-Integrated ECC w/ Torsion Points as Black Box Alternatives in Aircraft Security [🔗](#)

2025

Soft Computing Applications in Modern Power and Energy Systems

Accepted

Certifications & Achievements

Certifications: Quantum Computing (Udemy, 2025) – covered **Python, Q#, Qiskit**; ML Specialization (Coursera, 2023); **Google Cloud** Career Practitioner & Foundation Programs (2023–24).

Achievements: **Top 25/548** in VITISH (SIH) Hackathon 2024; invited to pitch at VNEST incubator. Awarded **PMSS Scholarship** for academic excellence & leadership; **Open source contributor** - merged PRs in projects with **1K+ stars**.

Extracurricular

Volunteer Teacher - U&I Trust

July 2024 – Apr 2025

- Led academic sessions for underprivileged students; supported a **INR 2.13 crore** education fundraiser through outreach.